Designing a Hydroelectric Generator to Accommodate Various Water Sources

Data

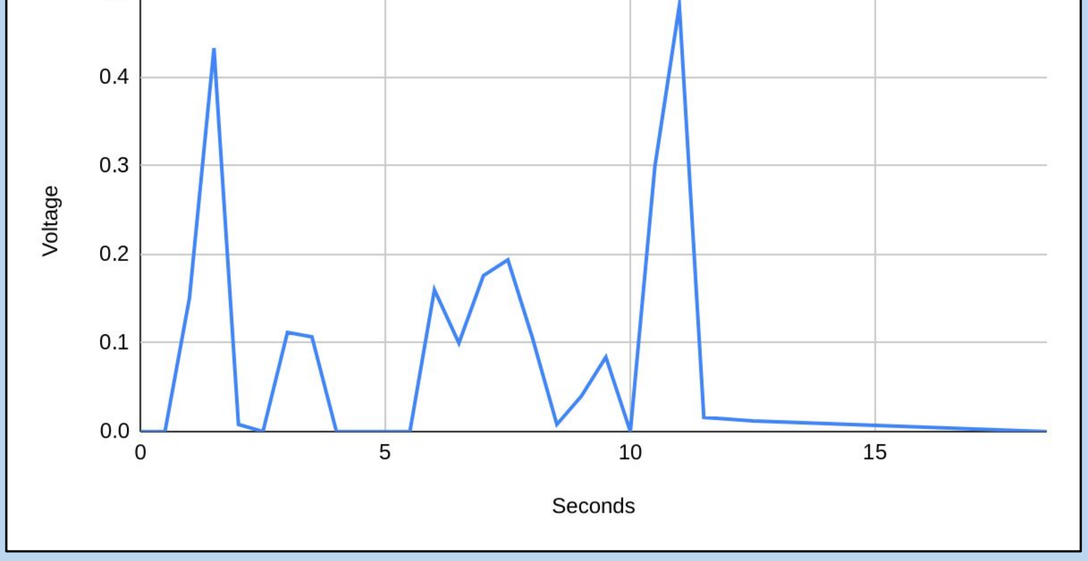
Flow Rate (seconds)

Trials	3.81cm (1.5 in) pipe	1.91cm (¾ in) pipe
1	11.58	30.9
2	13.15	31.93
3	13.96	40.01
4	12.73	35.09
5	11.68	32.84
6	10.68	34.15
7	12.23	34.62
8	11.96	40.34
9	10.59	35.39
10	11.71	38.69
11	12.33	37.23
12	11.26	38.12
13	11.36	37.97
14	10.78	37.59
15	11.06	37.44
16	12.84	31.63
17	11.65	36.44
18	11.26	32.72
19	10.97	38.2
20	11	29.23
21	10.84	39.24
22	11.54	37.22
23	11.96	31.04
24	12.52	30.82
25	12.05	35.06
26	10.7	35.32
27	12.98	37.38
28	11.97	29.3
29	12.93	38.56
30	10.89	35.5
Average Flow Rate	11.235	33.2
Standard Deviation	0.865	3.22

Voltage

Trials	3.81cm (1.5in) pipe Max Power	1.91cm (¾ in) pipe Max Power
1	1.301	0.599
2	1.979	0.404
3	1.316	0.481
4	1.685	0.572
5	4.24	0.461
6	1.297	0.477
7	1.623	0.415
8	1.452	0.391
9	1.367	0.386
10	2.899	0.602
11	1.306	0.51
12	1.311	0.542
13	1.428	0.517
14	1.806	0.566
15	1.297	0.573
16	1.55	0.355
17	1.935	0.606
18	1.15	0.349
19	1.739	0.398
20	1.918	0.432
21	1.444	0.483
22	1.528	0.612
23	1.737	0.597
24	1.115	0.626
25	1.212	0.645
26	3.567	0.384
27	4.239	0.388
28	1.104	0.636
29	1.456	0.57
30	1.072	0.372
Average Output	1.7831	0.4855

3.81cm (1.5in) Pipe Trial 5 Graph Progression



1.91cm (¾ in) Trial 3 Graph Progression

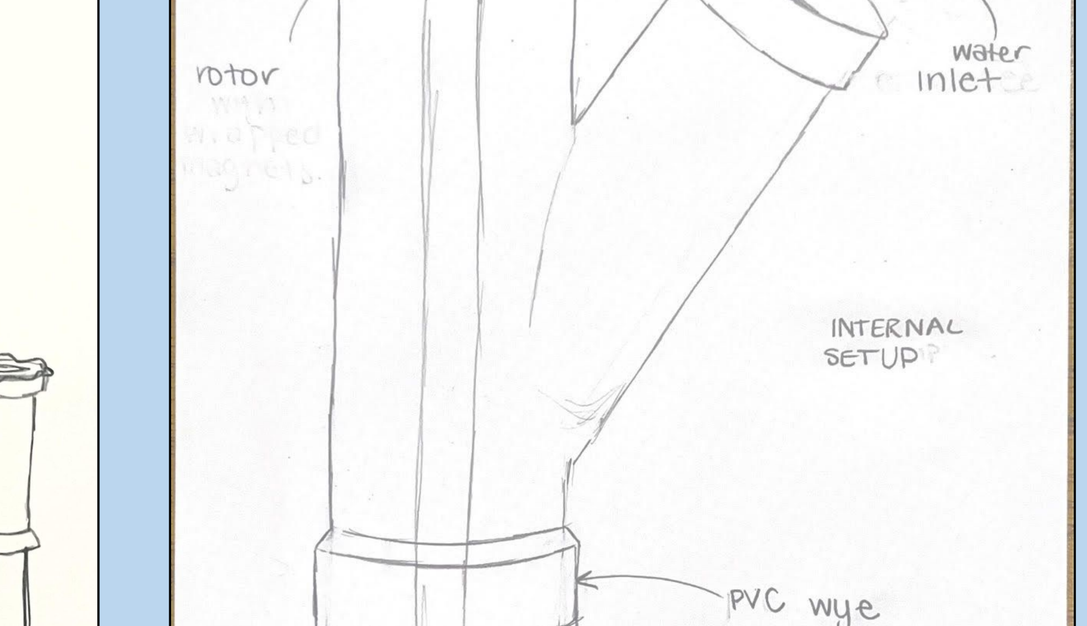
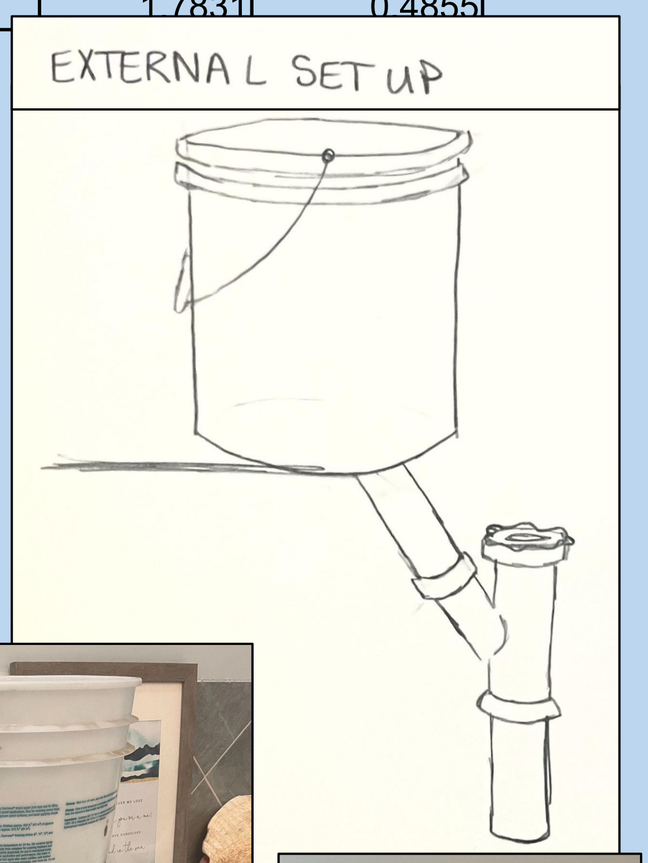


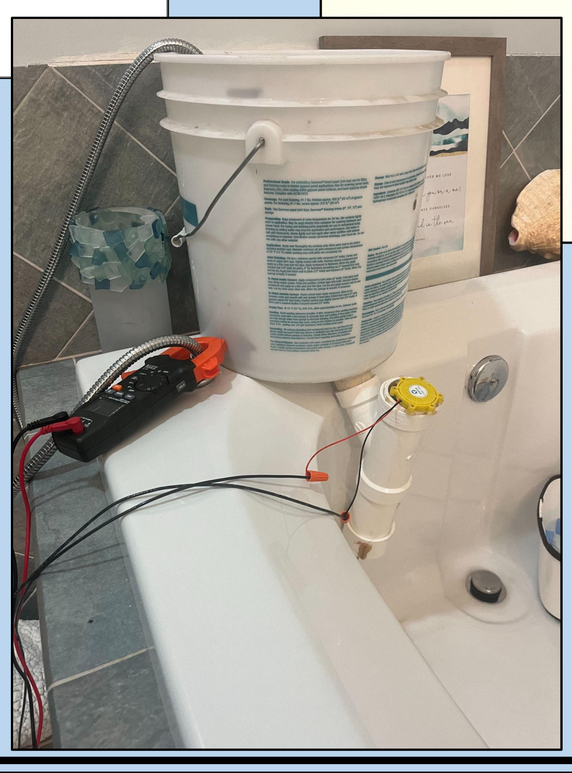
Table Key

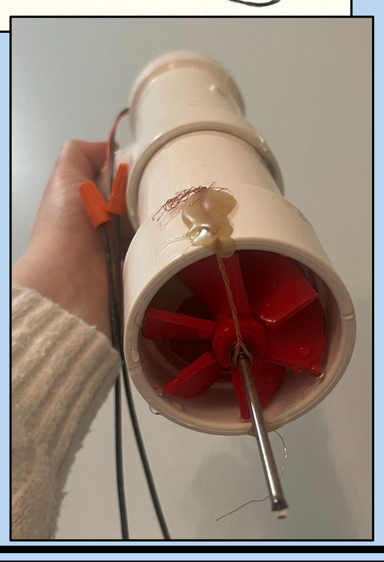
Flow Rate - Time (seconds) it takes to empty a 4 gallon Bucket

Voltage - Max voltage measured in one bucket cycle









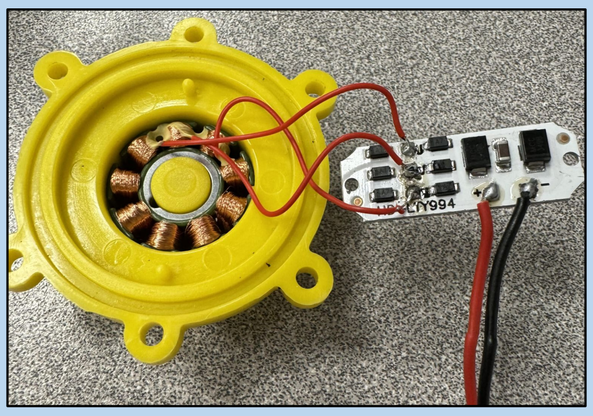
Problem

- Access to electricity is not attainable off the grid
- Lack of safety without proper communication
- Hydropower usually requires heavy water flow
- Often campers don’t stay near major water rapids
- Design a small hydroelectric generator that can accommodate slower water flow with less pressure

Hypothesis

By using a large bucket that empties through a small pipe, the head pressure created can be used to adapt slower water flow to produce sufficient electricity generated by a small-scale hydroelectric generator.

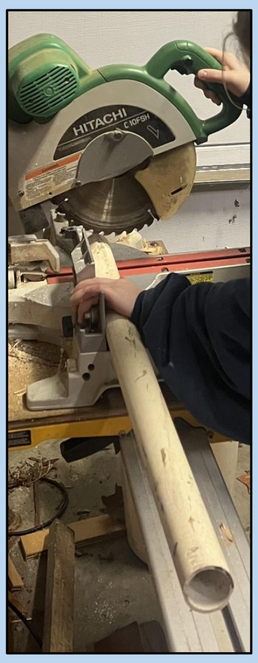
Stages Of Development











Final Design





Results

- The flow rate out of the bucket with the wider pipe fitting had a larger flow rate on average with 11.235 seconds per 4 gallons
- The flow rate of the smaller pipe fitting was 33.200 seconds per 4 gallons on average
- The wider pipe produced 1.7831 volts on average
- The narrower pipe produced much less with 0.4855 volts on average
- The smaller pipe’s voltage varied more on the multimeter, while the wider pipe’s voltage per second was more consistent with one spike

Conclusion

Because the voltage was significantly different between the faster and slower water flow, the hypothesis was not supported.

Future Testing

Modify my design to produce less friction and in return output more significant voltage